Optimizing Resort Amenity Management through an In-House Centralized Booking Application

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This report was formatted by Liennys Fuentes Perez based on the collaborative research, system requirements, and data analysis provided by the project group.

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Business Case *(Prepared by Emily)*

Executive Summary

Guests at The Harlow Hotel are dissatisfied with the current systems to reserve sports equipment, cabanas and chairs at the pool and beach, and dining at the three restaurants on property. The current systems for the sports equipment and pool and beach seating work on a first-come-first-serve basis, requiring guests to go to these locations when they want to use these services. This creates frustration and makes it difficult to plan if something is not available when the guests arrive. The three restaurants are on OpenTable, however, recently, reservations have been double-booked and OpenTable's service fee is higher than the hotel would like.

To remedy these issues, we are proposing creating and implementing a hotel-owned booking app, where guests can book all of these experiences right from their phones. Real-time availability for all equipment, seating, and dining will be visible to the guests, allowing them to reserve when is most convenient to them. There will also be a time limit on pool and beach seating, as well as equipment that is of high demand, so more guests can enjoy it.

We expect the implementation of this new app to increase guest satisfaction and experience, lower risk of double-bookings at the restaurants, increase revenue, reduce third-party costs, and reduce labor. The estimated cost for this project is $ 70,000 with an annual cost of $25,000. One key risk for this project, especially at the start is that there may be some technical issues resulting in accidental canceled bookings. While the technical issues are being resolved, we will offer points if a problem like this occurs, that the guests can use towards other bookings or when purchasing anything on property.

Problem Statement

The current reservation processes for equipment rentals, and pool and beach seating is done manually by hotel staff. It relies on a first-come-first-serve basis that makes availability unpredictable and inconsistent. Dining reservations work the same, though, they can also be booked in advance through OpenTable. However, this has led to several double-bookings in recent months. All this causes guest frustration, demand management issues, high third-party fees, and ultimately leads to revenue loss. This also takes up too much of the hotel staff's time with having to take the reservations manually and deal with the issues this system causes. The hotel requires a reliable and easy-to-navigate reservation system.

Environmental/Context Analysis

The hotel currently relies on a manual first-come-first-serve system for its equipment rentals and pool/beach rentals. It also uses a third-party platform for its dining reservations. The current processes lead to operational issues, poor customer experience, increase in the staff's workload, and overlapped dining reservations.

Guests are now expecting a more modern approach to hotel service booking systems, that include the ability to book ahead of time. Other resorts are beginning to implement digital systems with similar features to improve and facilitate operations. In addition, OpenTable, as well as other third party dining reservation platforms, charge a service fee, and increase risks that are out of the hotel's control.

Due to the advancements in technology, including real-time availability tracking, digitizing bookings is entirely possible. Already, the hospitality business as a whole has integrated digital systems to improve processes.

The hotel is experiencing the consequences of its outdated reservation system through unfortunate guest feedback and system collapse. However, there is a way to improve this, increase customer ratings, better structure the staff's time, and stand out as a hotel, by integrating a more efficient and modernized system.

Objectives & Success Metrics

The main objectives of this project are to increase guest ratings and satisfaction, make the system more efficient and reliable, reduce double-booking risks, reduce third-party costs, better structure staff workload, increase service utilization, and increase revenue.

Success will be measured by guest satisfaction score improvements, a decrease in double-bookings and third-party costs, and an increase in hotel amenity usage.

Options Considered

Keep The Current System: The benefits of continuing with the current system are that there is no added cost, no need to train staff on the new system, and no resources diverted to a project. The cons of keeping the current system are that all of the issues remain. Meaning, customer satisfaction would continue to decline, hotel staff would still be wasting valuable time on manual reservations, and booking overlaps and general problems would remain. We did not choose this option because it does not solve the problems.

Improve The Manual System: This would mean creating a cleaner schedule/calendar that the staff would use when making guests' reservations. The benefits of this would be very little added costs and that it would be a quick and simple solution. The cons of this would be that it would still be a manual system, we would still be paying a third-party for dining and carry the risks of that, and the system would still be outdated and unpredictable. We did not choose this option because it is not a permanent solution.

Buy A Pre-Built System: This would mean buying a pre-built reservation system from a vendor. The benefits would include a digitized option for the guests, a quicker implementation process, and support from the vendor. The cons would include having to pay for this as long as we use it, and not being able to fully customize it to our specific needs. We did not choose this option because it does not fully meet our needs.

Design Our Own System: This option includes designing, building, and implementing our own system. The benefits of this include being able to fully customize it to the hotel's specific needs, reduce booking issues and third-party fees, being able to better allocate staff workload, and improve guest experience. The cons include the initial cost, as well as that it would take more time to complete. We chose this option because we believe it would solve our main problems.

Recommended Solution

The recommended solution is to design, build, and implement our own reservation system. This is the best option because it improves guest ratings, and reduces unnecessary costs and scheduling conflicts. It also allows for the hotel to better structure its staff to further improve customer experience.

The new app will include the ability to reserve sports equipment, cabana and chair seating, and dining. It will show guests real-time availability, allowing the guests to schedule their day and activities in a much simpler way. Since cabanas, chairs, and certain sports equipment are in high demand, each guest will have a limit to how much time they can rent the amenities for, so more guests can enjoy them.

Implementing this is expected to lead to higher guest satisfaction, lower costs, lower risks, increased amenity usage and revenue, and improved staff time allocation.

Financial Analysis

The upfront costs of this project include system development and design, new tablets for the staff, implementation, and staff training. Ongoing costs include any maintenance or updates needed throughout time, technical support/help, and security monitoring.

Tangible benefits of this project include reduced third-party fees, reduced labor costs, and increased revenue. Intangible benefits include higher online ratings, improved guest experience, and an additional unique hotel feature.

Though there is an initial cost for this project, in the long run, saving on unnecessary costs and compensation for booking issues, as well as the projected increase in amenity usage and revenue, gives the hotel a great return on investment.

Risk & Mitigation

Risk 1: Technical issues lead to accidental cancellations by the system. To mitigate this problem, the hotel would provide points that guests can use to pay anywhere on property, if a booking problem occurs. The guests will also be able to rebook their desired amenity for an available time.

Risk 2: Guests may not use the system very much (probably just at first). As a solution, the hotel would offer free points just for signing up on the app and booking services.

Implementation Plan

The implementation of the plan will follow a schedule with different phases. The phases include planning, analysis, design, implementation and maintenance. The planning phase will include the project scope, feasibility, resource estimates, identifying risks, creating an initial budget, and making the project plan. The analysis phase includes identifying the current system, identifying the problems, figuring out requirements by listening to staff and management, and reviewing the constraints. The design phase includes creating the system architecture and security features, designing how reservations are made, modified, and canceled, and creating prototypes to test. The implementation phase includes developing the system and integrating it into daily operations at the hotel, which includes training the staff. Finally, the maintenance phase includes fixing any issues that may arise, as well as updating the system throughout time.

In order to complete this project, we would need a team that includes a project manager, system designers, management of the hotel, an information security team and training staff.

The project is expected to take around 8 months. After development is complete, testing would commence, leading to a soft launch if all goes well. The guests will be notified via email of the new app, and support will be available to them if needed.

Preliminary Feasibility Analysis *(Prepared by Hannah)*

The proposed hotel owned booking application appears to be both economically and technically feasible for the hotel to implement. From an economic standpoint, the system is expected to provide several benefits to the hotel. Some tangible benefits include reducing double bookings, lowering third party reservations fees currently paid to OpenTable, reducing labor associated with manual reservation processes, and increasing revenue through more efficient reservation management. The system is also expected to provide several intangible benefits, these include improved guest satisfaction, a more convenient guest experience, increased operational efficiency, and improved communication between hotel departments.

Guests will benefit from being able to view real-time availability and reserve experiences in advance instead of relying on first-come-first-serve processes. This is expected to improve the overall customer experience and strengthen the hotel's reputation.The hotel is estimated to save approximately $18,000 annually in third party reservation fees and $12,000 annually in labor costs. In addition, the hotel is projected to generate approximately $20,000 in additional annual review from increased amenity and dining reservations.

The project will require both one time and recurring costs. Estimate one time costs include approximately $70,000 for application development, database integration, software licensing, employee training, testing and implementation. Estimated recurring annual costs for hosting, maintenance, cybersecurity updates, and technical support are projected to total approximately $25,000 per year.

The proposed system is also considered technically feasible because the required technology already exists and is commonly used throughout the hospitality industry.The system would use a centralized reservation database with real-time availability tracking for dining, sports equipment, cabanas, and pool and beach seating. Guests would access the system through a mobile or web platform, while hotel staff would have administrative access to manage reservations and monitor availability. Potential technical risk include temporary system downtime and booking errors during the initial launch phase; however, these risks can be minimized through system testing, employee training and ongoing technical support.

Overall, the proposed booking application is considered economically and technically feasible and is expected to improve hotel operations while enhancing the guest experience.

Stakeholder Analysis ***(Prepared by Luis)***

In order to identify and keep track of everyone who is impacted or can have an impact on the project, a stakeholder analysis has been created.

Stakeholder Analysis					
Role	**Contact name**	**Category**	**Power**	**Interest**	**Engagement Strategy**
Hotel General Manager	Sofia Reyes	Internal	High	High	**Manage Closely:** Involve heavily during the requirements gathering phase. Schedule bi-weekly executive briefings. Require formal sign-off on project scope, budget, and major milestones. Prioritize her feedback on ROI and change in overall guest experience.
Pool & Beach Operations Manager	Carlos Mendoza	Internal	Medium	High	**Active Consultation:** Involve heavily during the requirements gathering phase. Consult with app ideas to improve current complaints. Gather feedback during testing.
Restaurant operations manager	Aria Thompson	Internal	Medium	High	**Active Consultation:** Involve heavily during the requirements gathering phase. Conduct regular check-ins to ensure that the restaurant reservation system does not impede the restaurant's workflow. Consult on restaurant

Stakeholder Analysis					
					capacity and seating layout.
Guests Service Representative	David Kim	Internal	Low	High	**Provide Training:** During the apps designing and testing phase, provide a high amount of training and understanding of the apps systems and inner workings. In this manner he can train guests in need of help, gather feedback, and promote the usage of the app
Guests	(Not Specified)	External	Low	High	**Keep Informed:** Involve in the requirements gathering phase. Utilize surveys for a small subset during development to gain first-hand user feedback. Broadly communicate the app's benefits and instructions via marketing materials and front desk signage upon launch.
App Development Team Lead	Tony Stark	External	Low	High	**Task-Directed Communication:** Maintain daily or weekly agile stand-ups. Provide clear technical requirements, scope documents, and UI mockups. Keep the focus entirely on execution and removing development blockers.
Legal and Compliance	Saul Goodman	Internal	High	Low	**Keep Satisfied:** Engage only at mandatory approval gates. Submit data

Stakeholder Analysis					
					privacy policies, digital liability waivers, and ADA compliance reports for review well before the launch deadline to prevent launch blockage.

Requirements Gathering : Interviews *(Prepared by Emmanuel)*

Interviews were held to collect qualitative information from hotel employees who use the existing reservation system every day. Structured interviews, as opposed to a survey or observation, give interviewers the opportunity to ask follow-up questions. Responses can illuminate root problem causes instead of surface-level symptoms.

The intention was to gain an understanding of how the existing system affects day-to-day operations from various stakeholder standpoints. We also wanted to know what mattered to each stakeholder if we were to replace it. Four stakeholders from varying levels within the hotel were interviewed to help gain both executive and street level insight

Interview Participants

The following four hotel staff members were selected based on their direct involvement with the current reservation processes.

Participant Name	Title / Role	Interview Focus
Sofia Reyes	Hotel General Manager	Strategic priorities, budget concerns, high-level pain points, and project success criteria
Carlos Mendoza	Pool & Beach Operations Supervisor	Daily operational challenges with equipment rentals and pool/beach seating management
Aria Thompson	Restaurant Operations Manager	Dining reservation workflow, OpenTable issues, and double-booking incidents
David Kim	Guest Services Representative (Front Desk)	Guest-facing interactions, common amenity complaints, and front desk workflow limitations

All interviews were conducted in person at the hotel property, lasted approximately 30 to 40 minutes each, and were documented through handwritten notes and audio recordings with participant consent. Interviews were conducted individually to avoid anchoring bias between participants.

Interview Guide

I created a common set of seven questions that were asked in each of the four interviews to keep consistency among them all. Some minor tweaking of the questions occurred to align with the interviewee's area of responsibility (the amenity area that they manage) but the essence of each question stayed the same. The interview guide flowed from descriptive to evaluative to prescriptive questions (how is it done now? What are the problems? What would you want?).

Q #	Interview Question
Q1	Can you walk me through how guests currently reserve [pool/beach seating / sports equipment / dining]?
Q2	What are the biggest pain points or frustrations with the current system — for both guests and staff?
Q3	How much time do you or your team spend on manual reservation management each day?
Q4	What types of guest complaints do you receive most frequently regarding reservations or availability?
Q5	How do you currently track availability for seating, equipment, or dining tables?
Q6	If you could change one thing about the current reservation process, what would it be?
Q7	What features or capabilities would you most want to see in a new reservation system?

Interview Transcripts

Interview 1 — Sofia Reyes, Hotel General Manager

Question	Response
Q1: Current reservation process	Pool and beach seating relies on a paper clipboard maintained by pool staff. Restaurant reservations are managed through OpenTable, with a handwritten paper log at the host stand as a backup. Sports equipment operates entirely on a first-come,

Question	Response
	first-served basis; guests walk up to the rental desk and receive the next available item.
Q2: Biggest pain points	OpenTable fees run approximately $2,000 per month. We experienced three dining double-bookings in the past quarter, two of which generated negative online reviews. Guest satisfaction scores for amenities have declined noticeably over the past two quarters.
Q3: Time spent on manual management daily	Based on department head reports, at least two staff hours per shift — across all amenity areas combined are spent on manual reservation tasks: fielding availability questions, logging requests, and resolving complaints tied to the current system.
Q4: Most common guest complaints	The most frequent complaints are: beach chairs and cabanas being unavailable or already claimed by towels, difficulty securing dining reservations with confidence, and equipment rental queues with no way to estimate wait times.
Q5: How is availability currently tracked?	There is no centralized tracking system. The pool clipboard is the only tool for seating. OpenTable manages dining, but it does not integrate with our property management platform, so there is always a data gap.
Q6: One change to the current process	Eliminate our dependency on OpenTable and take full ownership of every reservation process in-house. We are paying a third party to create problems we then have to solve ourselves.
Q7: Desired features for a new system	A guest-facing mobile app with real-time availability for all amenities and dining. A staff dashboard for managing bookings.

Question	Response
	Time-limited reservations on high-demand items so more guests can access them throughout the day.

Interview 2 — Carlos Mendoza, Pool & Beach Operations Supervisor

Question	Response
Q1: Current reservation process	Guests walk to the pool deck and ask staff for an available chair. If one looks unoccupied, we assign it verbally and note it on a clipboard. Equipment is signed out at the rental desk using a paper log with the guest's name and room number. Return times are rough estimates only.
Q2: Biggest pain points	Guests arriving after 10 AM regularly find all chairs claimed, even when some are physically empty because someone left a towel to hold the spot. There is no way for guests to check availability before walking over. Equipment queues form as early as 8:30 AM. I spend the majority of my shift managing complaints rather than running pool operations.
Q3: Time spent on manual management daily	Roughly three hours per day are consumed by chair management disputes and equipment complaints. That is time I am not spending on water safety, service quality, or staff supervision.
Q4: Most common guest complaints	The top question every day is: 'Can I reserve a chair or kayak in advance?' — and the answer is always no. Guests also ask how long until a kayak is free, and I have no way to answer that with a paper log.

Question	Response
Q5: How is availability currently tracked?	A paper sign-out sheet at the equipment desk. For chairs, it is purely visual staff walking the deck to assess what is open. There is no digital tracking of any kind.
Q6: One change to the current process	A digitally enforced time-limit system for chair and equipment reservations. Guests should be able to book a kayak slot in advance, and there should be a hard cutoff so no single guest monopolizes equipment for a full day.
Q7: Desired features for a new system	Real-time inventory tracking for all equipment. Time-limited reservations for cabanas and chairs with digital enforcement. Staff alerts when equipment is overdue. A guest mobile app to book from their phone before walking to the pool.

Interview 3 — Aria Thompson, Restaurant Operations Manager

Question	Response
Q1: Current reservation process	Reservations come through OpenTable as the primary channel, via phone calls that staff log manually, and as walk-ins. Phone reservations are supposed to be entered into OpenTable, but this does not always happen, which is how double-bookings occur.
Q2: Biggest pain points	We have had three confirmed double-bookings in the past 60 days. Two of those guests left visibly upset; one published a one-star review online. OpenTable also charges a per-cover fee of approximately $1.50 per reservation-sourced diner, totaling roughly $2,000 per month.

Question	Response
Q3: Time spent on manual management daily	Managing reservations takes the host staff approximately 1.5 hours each evening. When a double-booking incident occurs, resolution takes an additional 10 to 15 minutes per table, plus manager involvement and potential service compensation for the affected guest.
Q4: Most common guest complaints	Double-bookings and not being seated at their reserved time. Guests also complain about hold times when calling to book, and about uncertainty over whether a desired time slot is actually available.
Q5: How is availability currently tracked?	OpenTable is the primary system, with a handwritten paper backup. The two are not synchronized in real time. That gap is the direct cause of every double-booking we have experienced.
Q6: One change to the current process	Full ownership of the reservation system in-house. One system, one source of truth, no third-party fees, and no external failure points that we cannot control.
Q7: Desired features for a new system	An in-house restaurant reservation platform with real-time table availability, automatic booking confirmations sent to guests, a digital waitlist, and a staff-facing dashboard that completely replaces the paper backup.

Interview 4 — David Kim, Guest Services Representative

Question	Response
Q1: Current reservation process	When guests at check-in ask about pool chairs or kayaks, we call the pool deck or physically walk over to check. For dining, we

Question	Response
	direct them to OpenTable or call the restaurant host stand. There is no single source of information at the front desk.
Q2: Biggest pain points	Guests check in excited about the amenities and we cannot give them accurate availability information. We cannot confirm whether a kayak is available right now, or whether tonight's dinner at the Italian restaurant has open tables. That creates a poor first impression at the moment it matters most.
Q3: Time spent on manual management daily	I estimate 30 to 45 minutes per shift spent answering amenity availability questions that I cannot fully answer. That time comes at the expense of check-ins, checkouts, and other guest service priorities.
Q4: Most common guest complaints	The most common complaint at the front desk is 'I could not get a chair' or 'I could not book dinner.' Guests also ask multiple times per week why the hotel does not have an app for this.
Q5: How is availability currently tracked?	We have no availability system at the front desk. We rely entirely on phone calls to the pool deck or restaurant host stand, both of which have their own limitations and are not always reachable during busy periods.
Q6: One change to the current process	A real-time dashboard at the front desk showing availability for all amenities and dining, no phone calls needed. Ideally, guests manage their own bookings through an app before they even need to ask us.
Q7: Desired features for a new system	A hotel-branded mobile app guests can use from the time they book their stay through checkout. Real-time availability for all

Question	Response
	amenities, instant booking confirmation, and a staff-facing view so we can assist guests who need help.

Concluding Thoughts

Five themes recurred in every interview, regardless of role or department:

- Respondents have no visibility into amenity availability while in real-time: Each respondent mentioned that staff nor guests have the ability to look up amenity availability without walking to a physical location or calling another department. This was the single most common problem mentioned across all interviews.
- Hours lost each day to manual reservation management: Each participant discussed how much labor time is lost every day to reservation activities between 30 minutes and 3 hours per shift that would be automated or eliminated with a digital solution.
- OpenTable causes financial bleed and double bookings: Both the General Manager and Restaurant Manager independently mentioned OpenTable's per- cover fees (~$2,000/mo) and risk of double-booking reservations as justification for why replacing their current system with a DIY solution made the most business sense.
- Guests consistently ask for ability to reserve in advance: Guests have expressed interest in being able to reserve amenities and dining in advance. The current system does not allow them to do this for seating or equipment whatsoever.
- Limits are required on reservations for high-demand equipment: Reservation time limits were mentioned by multiple respondents as being operationally necessary (as opposed to optional) to ensure all guests have access to chairs, cabanas, and other popular equipment.

System Requirements Document *(Prepared by Liennys)*

To solve the reservation issues identified and stay within the budget, all while giving our front-line teams the exact tools they asked for in the interviews, the new application must meet the following functional and nonfunctional design requirements:

Functional Requirements (What the app must do)

1. **The All-In-One Booking Engine**
    - The app must allow verified guests to view and book slots across all three categories: Dining Tables, Sports Equipment, and Seating (Chairs/Cabanas).
    - Bookings must link directly to a valid Guest Room ID and only be accessible between their check-in and check-out dates to protect the system's integrity and keep everything organized.
2. **Real-Time Syncing & Double-Booking Protection**
    - The system must use a single centralized cloud database so all devices and staff dashboards see the exact same availability at the exact same second.
    - The moment a guest selects a time slot or table the system must instantly "freeze" it for that user during checkout to guarantee it cannot be double-booked by anyone else.
3. **Smart Alerts & Notifications**
    - The app must instantly send automated booking confirmations and cancellation receipts via push notifications and email.
    - The system must automatically flag a piece of sports gear on the staff dashboard the minute a rental period expires without a return.
4. **Digital Boundary Enforcer (Fair Play Rules)**
    - The app must automatically enforce maximum rental windows on high-demand gear and pool seating, that way if a guest does not check into their reserved beach chair within a set grace period, the system must automatically cancel the booking and release the chair back to the open pool.
5. **The Staff Override Dashboard**

- The system must provide a dedicated interface for hotel staff, front desk clerks, and managers as staff must be able to manually book walk-in guests, override time limits when necessary, and instantly credit a guest's account with compensation points if an early launch glitch occurs.

Nonfunctional Requirements (How well it must do it)

1. **Security & Privacy**
    - All guest data, room numbers, and transaction information must be fully encrypted from end to end to ensure no violations of privacy.
    - The system must use role-based access control, meaning a restaurant host only sees dining tables while IT admins have full configuration access.
2. **User-Friendly & Mobile-First Design**
    - The guest interface must be completely responsive, looking and performing beautifully on both iOS and Android smartphones, as well as standard web browsers. Thus, user interface layout must be clean and intuitive enough for guests of all tech skill levels to use without needing instructions.
3. **Reliability & Uptime**
    - The system must maintain a 99.9% uptime rate so guests and staff are never left stranded without access to the database. To ensure so, the system must include an automated cloud backup failover, if the local hotel internet hiccups, our booking data is safe and won't get lost.
4. **Speed & Performance Capacity**
    - The system must smoothly handle at least 500 active, simultaneous users without slowing down or lagging.
    - When a guest or staff member searches for availability, the app must populate the results in under 2 seconds.